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KD-trees

Practice Quiz • 20 min • 4 total points

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Introduction to Range Searching

1D Range Searching

KD Trees

Queries in KD-Trees

Video: Queries in KD-Trees

7 min

Practice Quiz: KD-trees

Started

Range Trees

Range Trees: Extensions

Programming: Orthogonal Range Searching

Wrap-Up Quiz

🎉 Congratulations! You passed!

Grade received 100% To pass 75% or higher

KD-trees

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1. Suppose we have a collection of points and a KD-tree on these points such that there exists a query that takes $O(\sqrt{n} + K)$ time, will all queries on this KD-tree take $O(\sqrt{n} + K)$ time?
Correct. On this KD-tree take $O(\sqrt{n} + K)$ time. 1 / 1 point
- ☐ Yes

☒ No

Help me practice

Let's chat
- ☒ Correct

☐ Submit your assignment

Correct, for example if a query region completely fits inside of a region which is represented by an internal node of the KD-tree then we only traverse one path in the KD-tree which takes $O(\log n)$ time.

Try again
- Receive grade

To Pass 75% or higher

100%

View Feedback

We keep your highest score

1 / 1 point
2. In the KD-tree queries we have used rectangular regions as the query region. Can the query algorithm also be used with triangular regions?

☐ No, this is not possible.

☐ Yes, but we need to adapt the query algorithm.

☒ Yes, even the query algorithm stays the same.

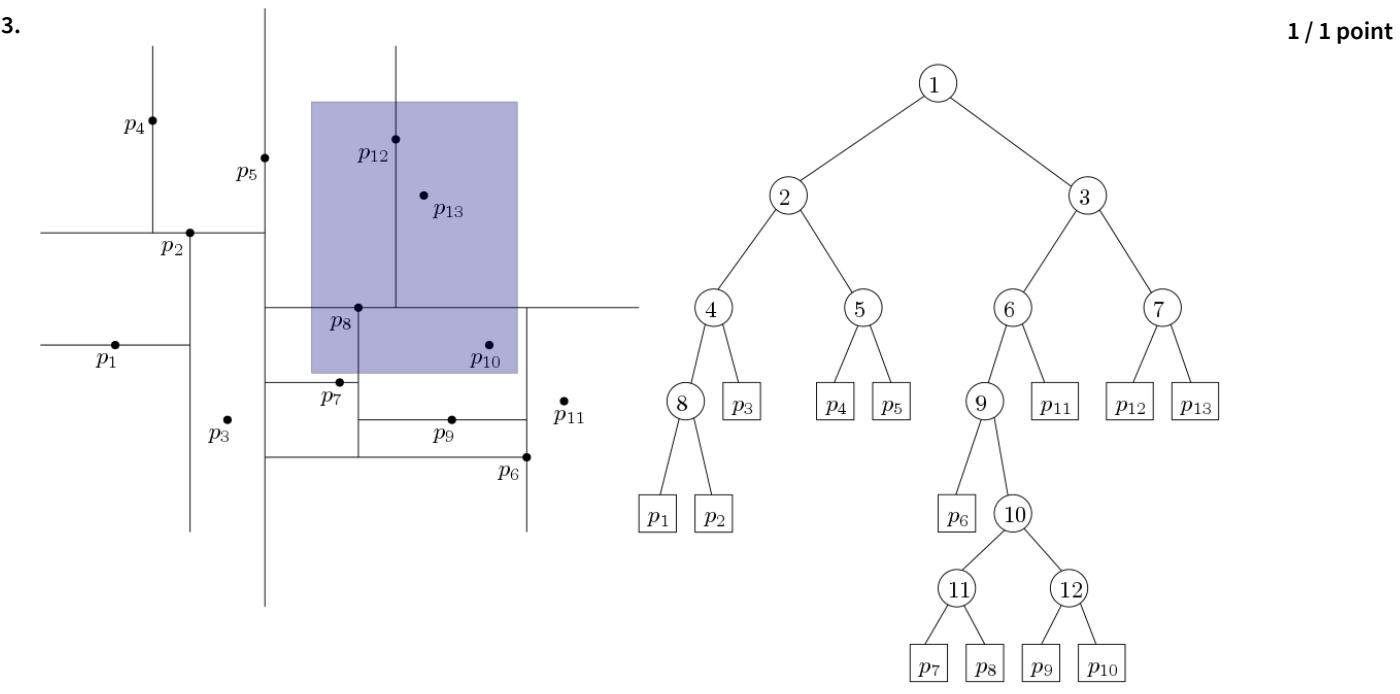
☒ Correct

☐ Like

☐ Dislike

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Correct, in each internal node the algorithm checks whether the area represented by the node intersects the query region. This would still work if this region had another shape.



The image above shows a subdivision of a point set and the corresponding KD-tree. Suppose we do a query with the blue region, which nodes will be visited? When the region corresponding to a node does not intersect the query region, you can consider the node to be not visited.

☒ 1 3 6 7 9 10 11 12 p8 p10 p12 p13

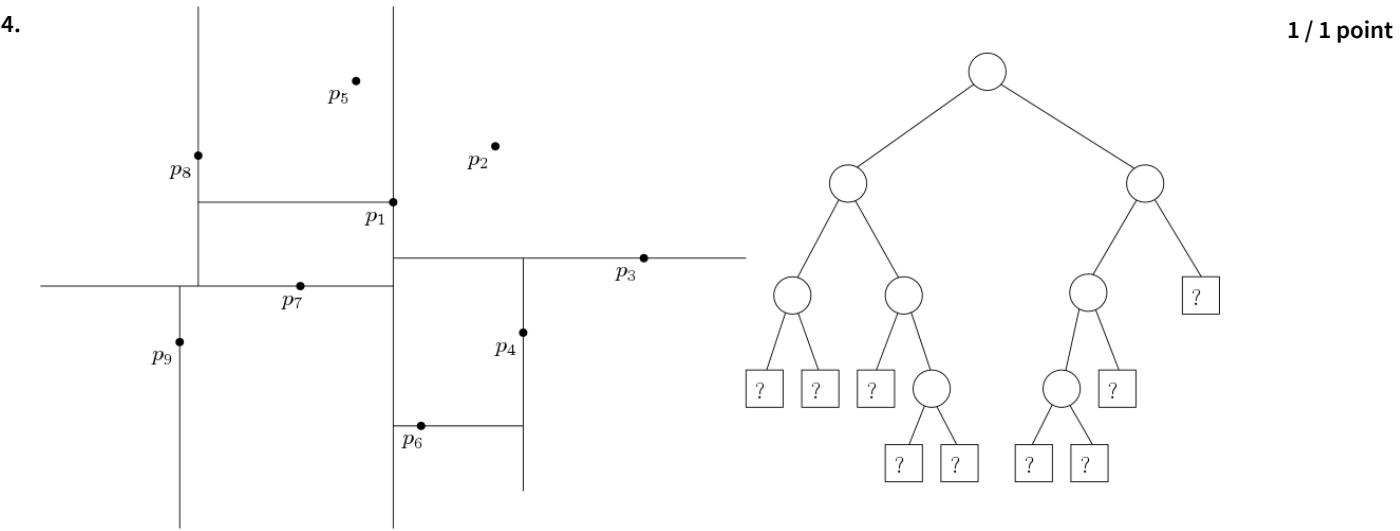
☐ 1 3 6 7 9 10 11 p6 p8 p10 p12 p13

☐ 1 2 6 7 9 10 11 12 p8 p10 p12 p13

☐ 1 3 6 7 9 10 11 p8 p10 p12 p13

☒ Correct

Correct.



Consider the subdivision and the corresponding KD-tree in the image above. What is the correct labelling of the leafs from left to right? You can assume that when we split a region, the upper/right sub-region are placed in the right sub-tree, and the lower/left sub-region are placed in the left sub-tree.

☒ p9, p7, p8, p1, p5, p6, p1, p3, p2

☐ p1, p2, p3, p1, p5, p6, p7, p8, p9

☐ p9, p4, p1, p2, p2, p5, p1, p5, p8, p7

☒ Correct

Correct.